

Unit 34: Indeterminate Forms and L'Hôpital's Rule — Full Solutions

Warm-up

1. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

Plug in: $\frac{4-4}{0} = \frac{0}{0}$ — indeterminate. Apply L'Hôpital:

$$\lim_{x \rightarrow 2} \frac{2x}{1} = 2(2) = 4$$

Answer: 4.

2. $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1$$

Answer: 1.

3. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = 1$$

Answer: 1.

4. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 3} \frac{2x}{1} = 6$$

Answer: 6.

5. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{\sin x}{1} = \sin 0 = 0$$

Answer: 0.

$$6. \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 1} \frac{3x^2}{1} = 3$$

Answer: 3.

Standard

$$7. \lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$$

$\frac{\infty}{\infty}$. Apply L'Hôpital:

$$\lim_{x \rightarrow \infty} \frac{6x + 2}{10x}$$

Still $\frac{\infty}{\infty}$. Apply again:

$$\lim_{x \rightarrow \infty} \frac{6}{10} = \frac{3}{5}$$

Answer: $\frac{3}{5}$.

$$8. \lim_{x \rightarrow 0} \frac{x^2 + 3}{x + 1}$$

Plug in directly: $\frac{0+3}{0+1} = \frac{3}{1} = 3$. This is NOT an indeterminate form — it's just an ordinary number. L'Hôpital's Rule does not apply here.

Answer: 3 (found by direct substitution, no L'Hôpital needed).

$$9. \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{\sec^2 x}{1} = \sec^2(0) = 1$$

Answer: 1.

$$10. \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$\frac{\infty}{\infty}$. Apply L'Hôpital:

$$\lim_{x \rightarrow \infty} \frac{1/x}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Answer: 0.

$$11. \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{3x}$$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{2e^{2x}}{3} = \frac{2e^0}{3} = \frac{2}{3}$$

Answer: $\frac{2}{3}$.

$$12. \lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$$

$\frac{0}{0}$ (since $\sin \pi = 0$). Apply L'Hôpital:

$$\lim_{x \rightarrow \pi} \frac{\cos x}{1} = \cos \pi = -1$$

Answer: -1 .

Challenge

$$13. \lim_{x \rightarrow 1} \frac{x^3 - 1}{4x^3 - x - 3}$$

Check: numerator = $1 - 1 = 0$; denominator = $4 - 1 - 3 = 0$. $\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 1} \frac{3x^2}{12x^2 - 1} = \frac{3(1)}{12(1) - 1} = \frac{3}{11}$$

Answer: $\frac{3}{11}$.

$$14. \lim_{t \rightarrow 0} \frac{\sin(t^2)}{t^2}$$

Check: numerator = $\sin(0) = 0$; denominator = 0. $\frac{0}{0}$. Apply L'Hôpital (differentiating with respect to t):

$$\lim_{t \rightarrow 0} \frac{\cos(t^2) \cdot 2t}{2t}$$

Cancel the $2t$ (valid as $t \rightarrow 0$ but $t \neq 0$):

$$\lim_{t \rightarrow 0} \cos(t^2) = \cos(0) = 1$$

Answer: 1.

$$15. \lim_{x \rightarrow 1} \frac{x-1}{\ln x - \sin(\pi x) + x - 1}$$

Check: numerator = 0; denominator = $\ln 1 - \sin \pi + 1 - 1 = 0 - 0 + 0 = 0$. $\frac{0}{0}$. Apply L'Hôpital:

Numerator's derivative: 1.

Denominator's derivative: $\frac{1}{x} - \pi \cos(\pi x) + 1$.

At $x = 1$: $\frac{1}{1} - \pi \cos(\pi) + 1 = 1 - \pi(-1) + 1 = 1 + \pi + 1 = 2 + \pi$.

$$\lim_{x \rightarrow 1} \frac{1}{\frac{1}{x} - \pi \cos(\pi x) + 1} = \frac{1}{2 + \pi}$$

Answer: $\frac{1}{2 + \pi}$.

$$16. \lim_{x \rightarrow 0} \frac{\tan^2 x (1 - \cos x)}{x^4}$$

Rather than brute-forcing repeated L'Hôpital on this tangled product, it's much cleaner to split the limit into two recognizable pieces:

$$\frac{\tan^2 x (1 - \cos x)}{x^4} = \left(\frac{\tan x}{x} \right)^2 \cdot \frac{1 - \cos x}{x^2}$$

First piece: from Problem 9, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$, so $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^2 = 1^2 = 1$.

Second piece: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$. Check: $\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{\sin x}{2x}$$

Still $\frac{0}{0}$. Apply again:

$$\lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2}$$

Combine both pieces:

$$1 \times \frac{1}{2} = \frac{1}{2}$$

Answer: $\frac{1}{2}$.

17. $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$$

Still $\frac{0}{0}$. Apply again:

$$\lim_{x \rightarrow 0} \frac{\sin x}{6x}$$

Still $\frac{0}{0}$. Apply a third time:

$$\lim_{x \rightarrow 0} \frac{\cos x}{6} = \frac{1}{6}$$

Answer: $\frac{1}{6}$.

Applied

18. Show $\lim_{t \rightarrow a} \frac{t^2 - a^2}{t - a} = 2a$.

Check: at $t = a$, numerator $= a^2 - a^2 = 0$; denominator $= a - a = 0$. $\frac{0}{0}$. Apply L'Hôpital (differentiating with respect to t , treating a as a constant):

$$\lim_{t \rightarrow a} \frac{2t}{1} = 2a$$

Confirmed. This exactly matches the difference-quotient definition of the derivative from Unit 7: for $f(t) = t^2$, this limit is precisely $f'(a)$, and since $f'(t) = 2t$, we get $f'(a) = 2a$ — the same answer, arrived at two different ways.

19. $C(x) = x^2 + 3x$, evaluate $\lim_{x \rightarrow 0} \frac{C(x) - C(0)}{x}$.

$C(0) = 0$. This limit is $\frac{0}{0}$ at $x = 0$. Apply L'Hôpital (differentiating the numerator as a function of x , since $C(0)$ is just a constant):

$$\lim_{x \rightarrow 0} \frac{C'(x)}{1} = C'(0)$$

$C'(x) = 2x + 3$, so $C'(0) = 3$.

Answer: 3, which matches $C'(0) = 3$ exactly — confirming that this limit is the definition of the derivative at $x = 0$.

20. Show $\lim_{t \rightarrow 0} \frac{1 - e^{-kt}}{t} = k$.

Check: at $t = 0$, numerator $= 1 - e^0 = 1 - 1 = 0$; denominator $= 0$. $\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{t \rightarrow 0} \frac{ke^{-kt}}{1} = ke^0 = k$$

Confirmed: the limit equals k .

21. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)}$

$\frac{0}{0}$. Apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{3 \cos(3x)}{5 \cos(5x)} = \frac{3 \cos(0)}{5 \cos(0)} = \frac{3}{5}$$

Answer: $\frac{3}{5}$.